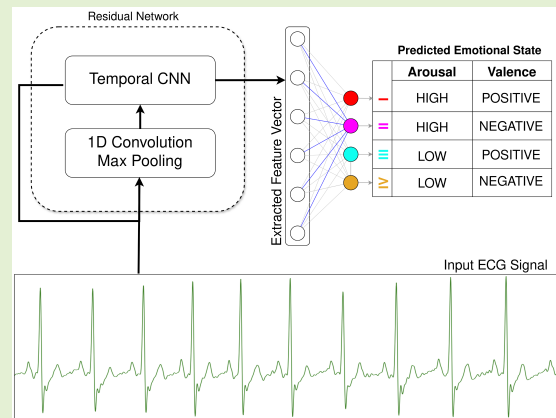


ECG-Based Automated Emotion Recognition Using Temporal Convolution Neural Networks

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Abstract—This study introduces a novel application of temporal convolutional neural networks (TCNNs) for automated emotion recognition (AER) using electrocardiogram (ECG) signals. By leveraging advanced deep learning (DL) techniques, our approach achieves the impressive classification accuracies of 98.68% for arousal and 97.30% for valence across two publicly available datasets. This methodology effectively preserves the temporal integrity of ECG signals, offering a robust framework for real-time emotion detection. Extensive preprocessing ensures high-quality input data, while cross validation confirms model generalizability. Our results demonstrate the potential of TCNN in enhancing human–computer interactions and healthcare monitoring systems through improved emotion recognition, paving the way for future applications in affective computing and wearable sensor technology.

Index Terms—Affective computing, biomedical signal analysis, convolutional neural networks (CNNs), deep learning (DL), electrocardiogram (ECG), emotion recognition, machine learning (ML), physiological signal processing, temporal convolutional networks (TCNs), wearable sensors.



I. INTRODUCTION

AUTOMATED emotion recognition (AER) [1] is the process of programmatically identifying and classifying affective responses to stimuli. In the context of interpersonal relationships, an individual's ability to recognize and reflect another's emotional state is known as empathy. The goal of AER is to mimic empathy in digital systems, with applications in improving human–computer interactions, behavioral monitoring, health, and wellness.

Emotions are multifaceted in their manifestation. In addition to the intangible and subjective psychological experience of “feelings,” an individual's emotional state has been shown to have quantifiable effects on their facial and vocal expressions, cognitive abilities and motor responses, and numerous physiological systems, such as heart rate, heart rate variability, skin temperature, perspiration, and respiratory rate [2], [3]. Assessing emotional state is a complex, multimodal, and autonomic process in humans, which, until recently, had been difficult

for computers to mimic. AER relies on the measurement of the physically observable characteristics of emotion using technologies, such as computer vision, behavioral monitoring of human–computer interactions, and measurement of physiological signals using biomedical sensors. Advancements in machine learning (ML) have made it possible to perform complex multimodal analyses of one or more of these characteristics, resulting in highly accurate classification of an emotional response [4].

Russell's circumplex model of human emotion [5] is used extensively in AER research for emotion classification problems. The model measures emotion in two dimensions: *arousal*, a measure of activation or excitement, and *valence*, a measure of pleasantness. Arousal may be *high* as in feelings of excitement or anger or *low* such as when feeling calm or tired. Valence may be *negative*, such as in feeling frustration or depression, or *positive*, such as feeling happy or contented. When plotted against each other, as shown in Fig. 1, the breadth of human emotion may be classified into specific quadrants, with each emotion belonging to a specific quadrant in the arousal/valence (A/V) space. Specific emotional states may be differentiated within each quadrant by considering the magnitude along each axis. For example, a feeling of excitement may be differentiated from one of delight or happiness based on the magnitude of the perceived state of arousal.

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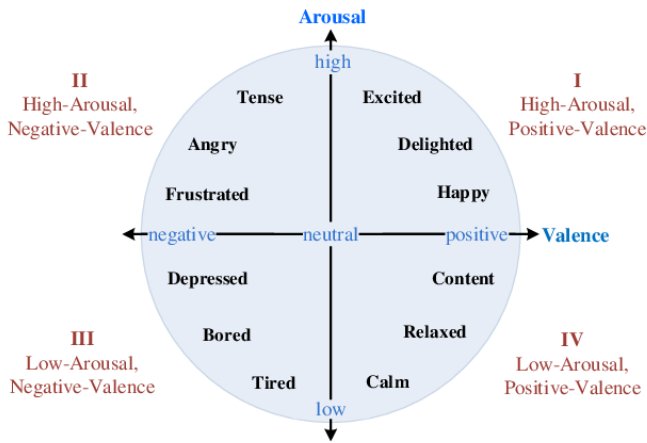


Fig. 1. Affective states in the valence–arousal space.

The traditional pipeline for AER begins with the capture of physiological signals, followed by signal preprocessing, and then manual feature engineering, followed by ML classification. Many traditional machine learning classifier design methods, e.g., k -nearest neighbors (KNNs), naive Bayes (NB), and support vector machines (SVMs), have been used with modest levels of success. Koelstra et al. [6] performed manual feature extraction of 14-node electroencephalogram (EEG) signals and an NB classifier, achieving 62.0% accuracy for arousal classification and 57.6% accuracy for valence classification. Subramanian et al. [7] repeated the study performed by Koelstra using a two-channel electrocardiogram (ECG), demonstrating comparable results with a classification accuracy of 60% and 59% for valence and arousal, respectively. Wiem and Lachiri [8] suggested an SVM classifier with statistical features extracted from ECG signals. The SVM classifier achieved the accuracies of 64.23% for arousal and 68.75% for valence.

In recent years, notable advances have been made using deep learning (DL) models for automatic feature extraction from physiological signals. As DL models are capable of performing simultaneous feature extraction and classification, they have the advantage of simplifying the overall AER pipeline. Santamaria-Granados et al. [9] propose the use of convolutional neural networks (CNNs) to ECG and galvanic skin response (GSR) signal data and achieves a classification accuracy of 71% for arousal and 75% for valence. CNNs have also been successfully applied in advancing research in biomedical fields other than emotion recognition. For example, in medical imaging, CNNs have been utilized for the detection of periodontal bone loss, demonstrating the flexibility and effectiveness of these models in processing and analyzing complex biomedical data [10].

Researchers have improved upon these results by adding additional structural analysis to the traditional CNN. Harper and Southern [11] propose to append a long short-term model (LSTM) for time-series analysis to CNN. They achieve 90% classification accuracy for high/low valence emotions against the same dataset provided by Santamaria-Granados. Hammad and Monkaresi [12] propose a feature fusion technique that utilizes parallel extraction of temporal and spatial characteristics using 1-D and 2-D CNN (PETSFCNN). They achieve

97.56% and 96.34% accuracy for arousal and valence, respectively. A recent study by Oğuz et al. [13] demonstrates the effectiveness of automated feature engineering of ECG signals for AER, achieving 83.61% accuracy for arousal and 78.28% for valence using bidirectional LSTM alone. This supports the use of automated feature engineering and CNN with structural augmentation.

In this article, we demonstrate a novel application of temporal CNN (TCNN) to AER problems to improve the state-of-the-art classification accuracy of ECG-based affective states. TCNNs have previously been demonstrated in GANs for the generation of audio signal data [14], [15]; however, when applied to physiological signal data, such as ECG, they have the benefit of maintaining structural integrity during convolution without the need to add additional models to the CNN as done in prior work. We demonstrate the classification accuracies of 98.68% for arousal and 97.37% for valence, validated using threefold leave-one-out cross validation and cross-dataset validation.

Contributions of This Article:

- 1) *Introduction of TCNN for AER*: First application of TCNN to ECG-based AER.
- 2) *High Classification Accuracy*: Achieving state-of-the-art accuracy for arousal and valence using two datasets.
- 3) *Comprehensive Data Preprocessing Pipeline*: Enhancing ECG signal quality using the Neurokit2 Python Toolkit.
- 4) *Cross-Dataset Validation*: Validating model generalizability across multiple datasets.

II. DATA PREPARATION

A. Dataset Descriptions

We use two publicly available datasets: ASCERTAIN [7] and DREAMER [16].

ASCERTAIN:

- 1) Includes ECG, EEG, GSR, and facial activity data from 58 participants watching 36 movie clips.
- 2) ECG signals were recorded at 256 Hz with two measuring electrodes and one reference electrode.
- 3) Participants rated valence (−3 to +3) and arousal (0–6) post-viewing.
- 4) Data quality was assessed on a six-point scale.

DREAMER:

- 1) Comprises ECG and EEG data from 23 participants watching 18 movie clips.
- 2) ECG signals were recorded at 256 Hz with two measuring electrodes and one reference electrode.
- 3) Participants rated valence (1–5) and arousal (1–5) post-viewing.

B. ECG Data Preprocessing

ECG is characterized by a recurrent wave sequence of P, QRS, and T-waves associated with each heartbeat, as shown in Fig. 2(a). The QRS complex is the most apparent waveform within the signal. It reflects the electrical activity within the heart during ventricular contraction. The time of its occurrence and its shape provide much information about the state of the heart. Due to its characteristic shape, it serves as the basis

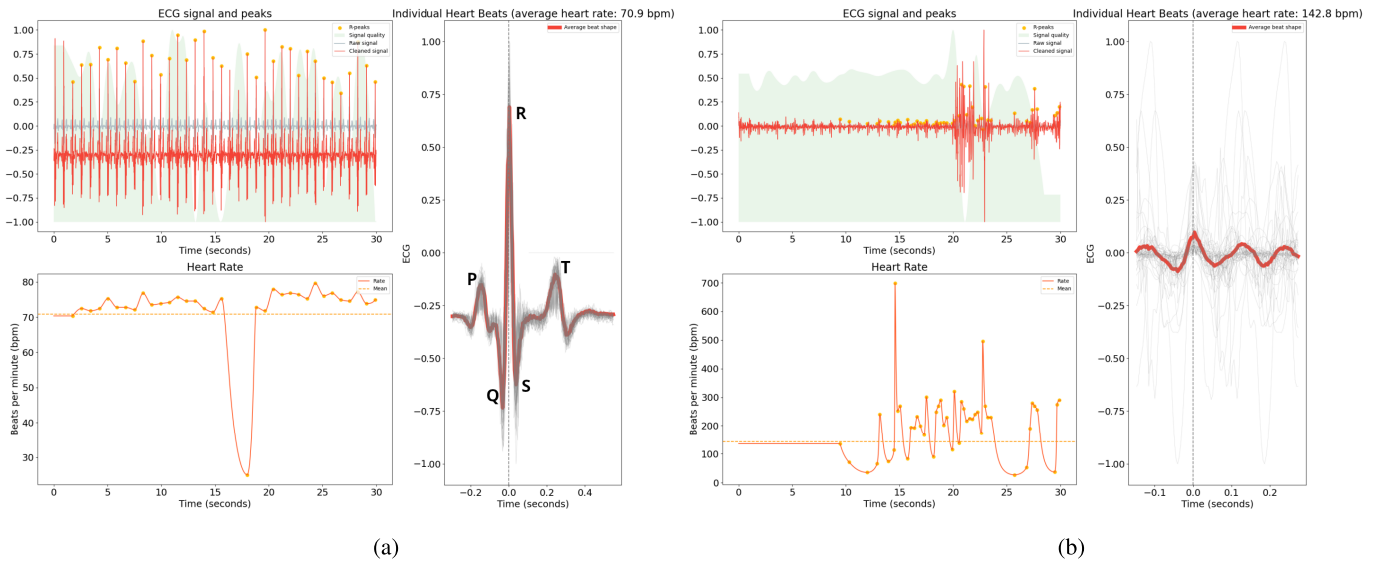


Fig. 2. Two ECG signals after preprocessing. (a) Acceptable ECG shows clear R-peak identification, consistent heart rate, and easily discernible QRS ECG wave. (b) Unacceptable ECG missing signal data, inconsistent R-peak identification, and erratic, invalid heart rate. No discernible QRS ECG wave.

for the automated determination of the heart rate as an entry point for analysis of the cardiac cycle. In that sense, QRS detection provides the fundamentals for almost all automated ECG analysis algorithms [17].

ECG signals are nonstationary, with nonlinear noise and artifact contamination having a time-dependent probability distribution. ECG signals are subject to a variety of noise, including baseline wander, power line interference, EMG signal interference, interference due to electrode movement, and signal inversion due to improper electrode placement [18]. We developed a custom preprocessing pipeline based on the open source Neurokit2 Python Toolkit [19] to clean and rank ECG signals for input into the proposed DL model.

The ECG preprocessing pipeline is made up of multiple steps. First, the signal is trimmed to the final 31 s, and the last second is discarded, resulting in a 30-s sample. ASCERTAIN and DREAMER trials are structured, so that the emotional response is evoked in the final 45–50 s of the video shown. However, we found that 30 s was sufficient for extremely high accuracy and significantly reduced training time. Next, the signal was checked for inversion due to misplaced electrodes and, if necessary, inverted again to correct. The signal is then rescaled to $[-1, 1]$ before applying a series of filters, including a 0.67-Hz high pass to eliminate baseline wander, a 50- or 60-Hz notch filter to remove ac powerline interference, and a median filter with kernel size 16, or 0.0625 s, to smooth out anomalous spikes in the data.

ASCERTAIN provides three-axis accelerometer data that allow correlation of EMG-related interference due to participant movement during the trial. For this study, we averaged the participant's motion in the three-axis space and zeroed out the ECG signal during periods when motion exceeded two standard deviations of the mean.

After the initial preprocessing is complete, r-peak analysis is performed to assess the quality of the ECG signal prior to use in the study. Samples with irregularly spaced R-peaks, irregular R-peak magnitudes, and samples demonstrating abnormally

high (above 150 bpm) or low (below 40 bpm) heart rates were discarded as being too noisy for proper evaluation. Fig. 2(a) shows an example of a clean, usable ECG after preprocessing. Note the consistent and evenly spaced R-peaks and clear average QRS complex shown. Fig. 2(b) shows an ECG discarded from the study due to high noise levels, inconsistent R-peak magnitude, an indiscernible QRS complex, and inaccurate heart rate measurements.

C. Affective Response and Personality Ratings

Each subject completed a self-assessment after each video clip, providing ground-truth data for the affective state in our dataset. ASCERTAIN uses a seven-point rating valence from -3 to $+3$ and arousal from 0 to 6. DREAMER uses a five-point scale rating both valence and arousal from 1 to 5. Valence and arousal scores for each trial are mapped to quadrants I, II, III, or IV in the A/V plane, as shown in Fig. 1. For this study, to avoid ambiguity, we omit data with neutral ratings for either valence or arousal from both datasets.

III. DEEP LEARNING MODEL

A. Temporal Convolution

Prior work has shown that feature fusion with structural analysis, be it spatial [9] or temporal [11], combined with CNN significantly improves prediction accuracy for AER problems. In our proposed model, rather than using a fusion approach with a structured model, such as LSTM or 2-D CNN, as used in previous work, we replace the traditional CNN with a TCNN with causal padding. This ensures that within each layer, the value at timestep T_i is only dependent on inputs from the previous layer at timesteps T_j where $j \leq i$, thereby preserving the time-dependent nature of the input signal.

Fig. 3 shows a TCNN with a kernel size of 2, using exponentially increasing kernel dilation with each layer, over n stacked layers. In this model, the output node at timestep T_i is dependent upon 2^n values from the original input, located at timesteps T_{i-2^n+1} through T_i . Stated another way, each output

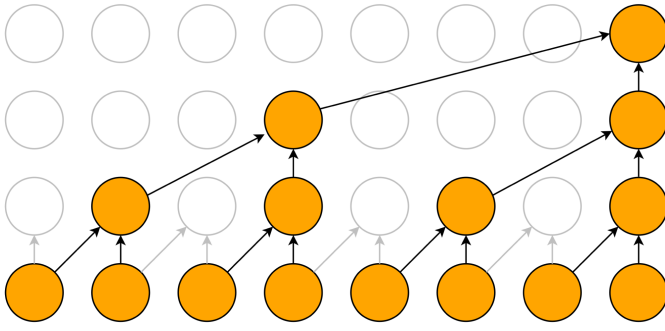


Fig. 3. TCNN with kernel size 2 and exponentially increasing dilation.

node has a receptive field size of $R = 2^n$. Given an input signal with sampling frequency F_s Hz, the receptive field size measured in seconds, R_{sec} , of each node in the output of an n -layer TCNN is given by

$$R_{\text{sec}} = \frac{1}{F_s} * 2^n.$$

B. Residual Blocks and Skip Connectors

The purpose of a residual block in CNNs is to facilitate the training of deep neural networks by addressing the vanishing gradient problem. The vanishing gradient problem occurs when the gradient signal becomes too small to be propagated back through many layers of a deep neural network during training, making it difficult to update the weights and biases in earlier layers.

A residual block also introduces a skip connection that bypasses one or more layers in a deep neural network. By passing information directly from an earlier layer to a later layer, the residual block allows gradients to flow more easily and helps prevent them from vanishing. This enables the network to learn more complex features and deeper representations and can lead to better performance [14]. The residual block achieves this by adding the input to the output of the convolutional layer within the block. This allows the network to learn the difference between the input and the output, rather than trying to directly learn the entire transformation.

C. Proposed Model

The proposed model is composed of a series of k residual blocks, as shown in Fig. 4. The block's input is first passed through a convolution and pooling layer, forming the foundation of a traditional convolutional autoencoder.

After pooling, TCNN with depth d is applied in a gated activation function, where each layer l has dilation 2^{l-1} and causal padding. We apply a gated nonlinear activation as used in WaveNet

$$z = \tanh(W_{f,k} * x) \odot \sigma(W_{g,k} * x)$$

where f and g denote the filter and gate, respectively, $\odot$ is the elementwise multiplication operator, and σ is a sigmoid function.

Finally, the blocks two outputs are produced: first, a $1 \times n$ convolution is applied to the output of the gated activation, where n is the desired length of the output feature vector,

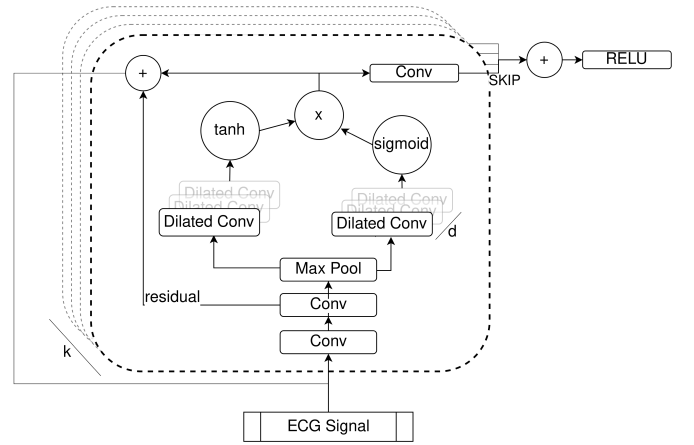


Fig. 4. Proposed model architecture.

to produce the skip connection output; second, the residual is added to the output of the gated activation, to produce the input to the next residual block in the sequence. After the final residual block has produced its output, the skip connections are summed, and a rectified linear activator is applied, producing the final extracted feature vector.

D. Emotion Classification

The final task is to classify the feature vector into a quadrant of the arousal/valence plane, as shown in Fig. 1. Each input signal is classified as either “high” or “low” in arousal and as either “positive” or “negative” in valence, which identifies its quadrant in the arousal/valence plane. The quadrant identifies the emotion’s valence as being positive or negative, and the level of excitement as being high or low, from which we can coarsely identify the general emotional response of the participant. Intuitively, we expect high accuracy from the model as a classifier, as transitions between affective states are highly correlated with changes in physiological responses. For example, increased arousal is highly correlated with increased heart rate. Classification is described in Section IV-A.

IV. TRAINING AND VALIDATION

A. Training Methodology

The proposed model is trained and evaluated using various combinations of samples from the ASCERTAIN and DREAMER datasets. Both datasets were split into training, validation, and test subsets, with the number of trials from each quadrant selected for each subset shown in Table I. Each trial contains a two-channel ECG recording, with each channel treated as a single sample during training and validation. The training, validation, and test subsets are split at the trial level, rather than the sample level; therefore, each ECG recording wholly contained within its subset. Validation and test subsets are balanced across all classes. During training, balanced batches are randomly selected from the training subset for each step.

Feature extraction training is performed with triplet loss using Manhattan, aka city block, distance with online semihard triplet mining. We use an Adam optimizer with an initial learning rate $\alpha = 1 \times 10^{-6}$. Classification accuracy is evaluated after each epoch against the validation dataset, and weights are

TABLE I

TRAIN, TEST, AND VALIDATION SPLITS. NO VALIDATION WAS PERFORMED WHILE TRAINING AGAINST DREAMER; THUS, ONLY TRAINING AND TEST SPLITS ARE DEFINED

ASCERTAIN Splits				
Quadrant	Total	Train	Validation	Test
I	705	645	15	15
II	648	588	15	15
III	464	404	15	15
IV	236	176	15	15
Total	2088	1968	60	60

DREAMER Splits				
Quadrant	Total	Train	Validation	Test
I	76	66	0	10
II	105	95	0	10
III	146	136	0	10
IV	87	77	0	10
Total	414	374	0	40

saved each time a new high accuracy score is reached. If the accuracy does not improve over 5000 epochs, the previous best weights are restored, and the learning rate is updated to be $\alpha = 0.8 * \alpha$. The model is allowed to train until accuracy has not improved in 50000 epochs. The final classification accuracy is evaluated against the unseen test set. Training and evaluation are performed using threefold leave-one-out cross validation with mean results reported in this article.

Classification accuracy is evaluated over a sample set using the model's current weights by computing the mean embedding for each class represented in the sample set. The predicted class for each sample in the sample set is the one with the shortest distance between the sample's embedding and the class' mean embedding. This method is utilized during training to evaluate the model's current state against the validation dataset and after training to evaluate the model's performance against an unseen test set.

In addition to threefold leave-one-out cross validation within each dataset, we cross validate our results by evaluating the classification accuracy against the DREAMER dataset after training the model on ASCERTAIN. This ensures that the reported results are not subject to overfitting.

B. Validation and Analysis

We evaluated the classification accuracy of the proposed model in three scenarios. Final results for a four-label classifier predicting quadrant within the A/V space is shown in Fig. 5. Separate confusion matrices for predicting arousal and valence independently, for each scenario, are shown in Fig. 6. Comparative results are summarized in Table II, and detailed classification performance metrics are shown in Table III. All reported classification metrics are the mean result for each scenario.

1) *Training and Validation on ASCERTAIN*: In the first scenario, the model is trained against the training subset of

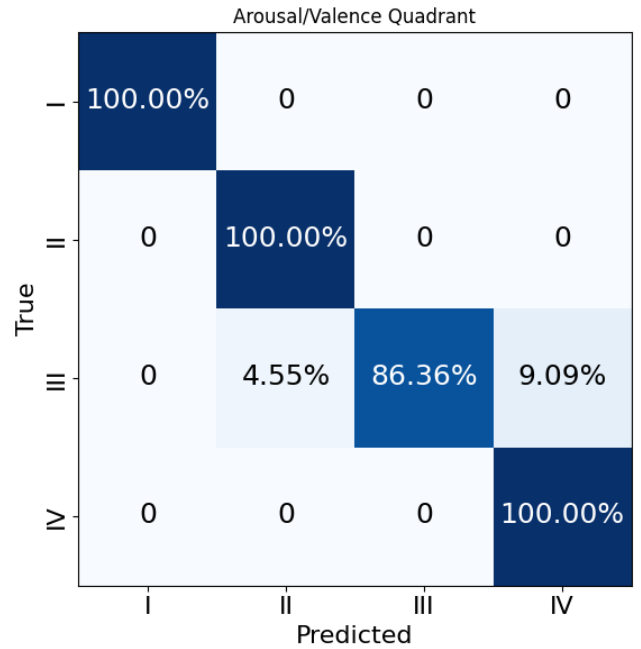


Fig. 5. Final classification into arousal/valence quadrants.

ASCERTAIN. During training, after each epoch, the model is evaluated using the validation subset from ASCERTAIN to assess classification accuracy and $f1$ score. After training has concluded, final classification accuracy and $f1$ score are calculated by evaluating the model against the test subset from ASCERTAIN. This was repeated three times, with new training, validation, and test splits generated for each iteration.

The model achieves the accuracy of 94.83% with an $f1$ score of 94.74% for arousal and 93.10% accuracy with an $f1$ score of 93.33% for valence. The mean results for predicting arousal and valence are shown in the confusion matrices in Fig. 6(a). Each confusion matrix is based on a total of 60 samples, with 15 samples from each quadrant. Four single-channel ECG samples were discarded during preprocessing yielding in 116 data points.

2) *Cross Validation Against DREAMER*: It is possible that the model may overfit due to the presence of trials from the same persons existing in each subset. For example, while each sample of the two-channel ECG recording of Person 1 watching Clip 1 may be in the test subset, other recordings from Person 1 watching different clips are likely to be present in the training subset. To ensure this was not the case, the model is reevaluated against the test subset selected from DREAMER, utilizing the weights from each training iteration against ASCERTAIN.

The model achieves 92.11% accuracy with an $f1$ score of 92.31% for arousal and 90.79% accuracy with an $f1$ score of 90.41% for valence, which shows that the model is generalizing well. The mean results are shown in the confusion matrices in Fig. 6(b). Each confusion matrix is based on a total of 40 samples, with ten samples from each quadrant. Four single-channel ECG samples were discarded during preprocessing yielding in 76 data points.

3) *Continued Training Against DREAMER*: Finally, we continue training against the DREAMER dataset, utilizing

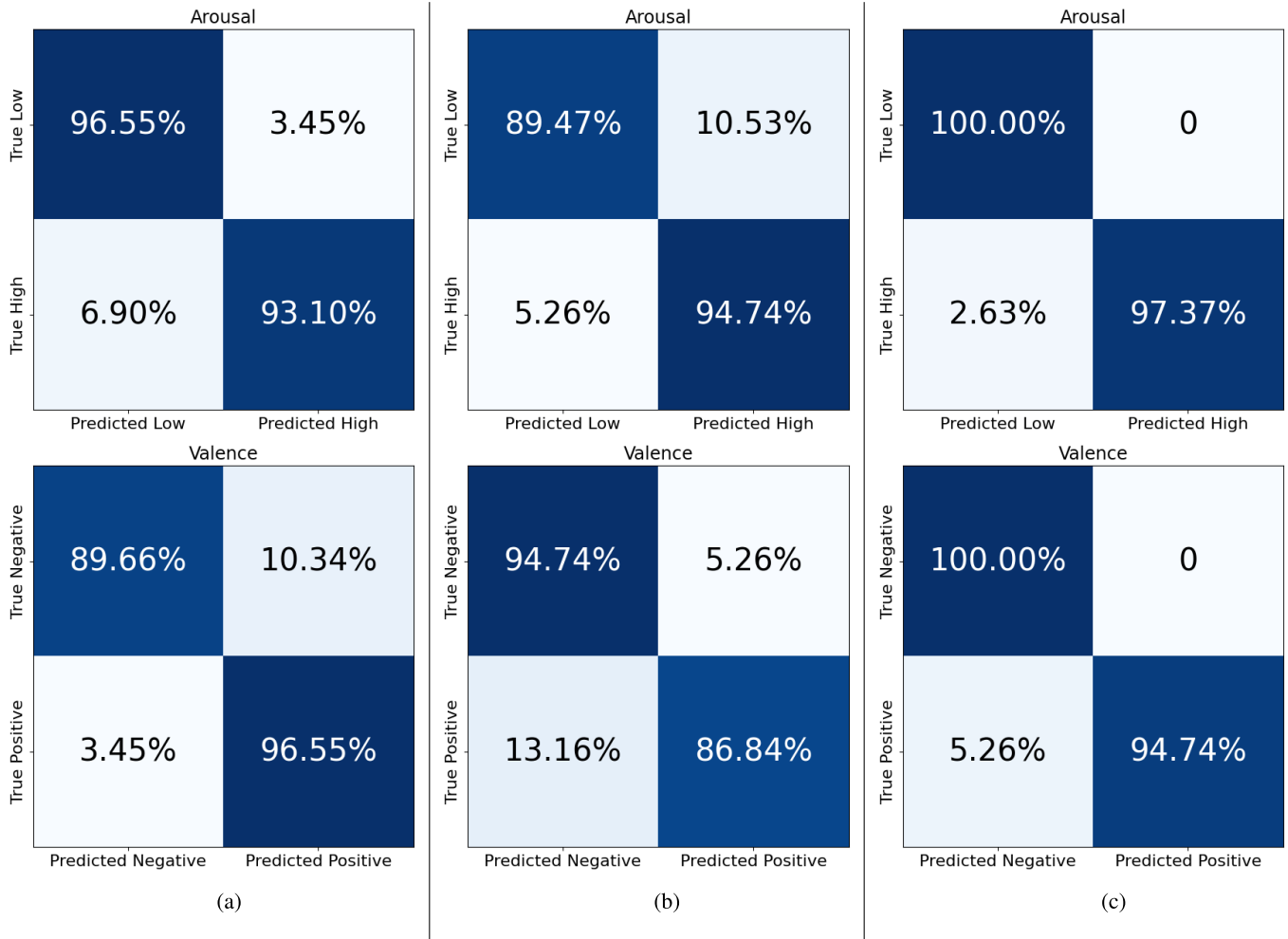


Fig. 6. Confusion matrices showing average performance predicting arousal and valence in three training scenarios. (a) Confusion matrix for training and validation on ASCERTAIN (Section IV-B.1). (b) Confusion matrix for cross validation against DREAMER (Section IV-B.2). (c) Confusion matrix for continued training against DREAMER (Section IV-B.3).

TABLE II
AER MODEL COMPARISON

	Training Dataset	Evaluation Dataset	Arousal Accuracy	Valence Accuracy
CNN+LSTM	AMIGOS	AMIGOS	-	86.0%
	DREAMER	DREAMER	-	90.0%
PETSFCNN	DREAMER	DREAMER	97.56%	96.3%
T-CNN	ASCERTAIN	ASCERTAIN	94.83%	93.10%
	ASCERTAIN	DREAMER	92.11%	90.79%
	ASCERTAIN + DREAMER	DREAMER	98.65%	97.30%

pretrained weights learned during the initial training on ASCERTAIN. During training, since no explicit validation set was selected from DREAMER, classification accuracy is evaluated against the test subset selected from ASCERTAIN. After training, classification accuracy is evaluated against the test subset selected from DREAMER. This was repeated three times, each with new training and test subsets generated from DREAMER.

Here, the model reaches 98.65% accuracy with an $f1$ score of 98.67% for arousal and 97.37% accuracy with an $f1$ score of 97.30% for valence. This improvement in performance is attributed to the 19.0% increase to the training dataset with the addition of DREAMER. Mean results are shown in the confusion matrices in Fig. 6(c). The predicted arousal and valence determine the predicted quadrant label for final classification within the arousal/valence space. The overall classification accuracy by quadrant, based on the results obtained in Section IV-B.3, is 96.0%, and the confusion matrix is shown in Fig. 5. Each confusion matrix in Figs. 5 and 6(c) is based on a total of 40 samples, with ten samples from each quadrant. Six single-channel ECG samples were discarded during preprocessing yielding in 74 data points.

V. DISCUSSION

A. Limitations and Interpretation of Results

The proposed TCNN model demonstrates significant improvements in the classification accuracy of both arousal and valence when compared with previous models, such as CNN + LSTM and PETSFCNN. Specifically, we achieved the accuracies of 98.68% for arousal and 97.37% for valence.

TABLE III
CLASSIFICATION PERFORMANCE METRICS

Scenario	# ECG	Arousal				Valence			
		Accuracy	F1-score	Sensitivity	Specificity	Accuracy	F1-score	Sensitivity	Specificity
Training and Validation on ASCERTAIN	116	94.83%	94.74%	93.10%	96.55%	93.10%	93.33%	96.55%	89.66%
Cross-Validation against DREAMER	76	92.11%	92.31%	94.74%	89.47%	90.79%	90.41%	86.84%	94.74%
Continued Training against DREAMER	74	98.68%	97.37%	100.0%	98.67%	97.37%	97.30%	94.74%	100.0%

These results highlight the effectiveness of TCNN in maintaining the structural integrity of ECG signals during convolution without the need for additional models, such as LSTMs, to handle temporal dependencies.

However, it is crucial to recognize the limitations of these findings. The test subsets used in our study are relatively small, which means that even a single misclassified sample can substantially impact the reported accuracy. Such small datasets may also result in overfitting during DL model training, which we address cross validation of two datasets. Furthermore, our results are highly sensitive to the quality of the ECG input signal. The preprocessing pipeline plays a pivotal role in determining the quality of the input data. While we employed robust preprocessing methods, the lack of detailed information about preprocessing steps in prior studies limits our ability to make a direct quantitative comparison. This sensitivity to preprocessing underscores the need for standardized preprocessing techniques in future AER research to ensure the reproducibility and comparability of results.

Although the results indicate superior performance of the TCNN model, it is premature to declare it the definitive approach based on reported numbers alone. When considering variability in test set selection and differences in ECG preprocessing, our model performs comparably to the PETSFCNN model. Nonetheless, the reliance on time-series analysis alone rather than on feature fusion with spatial transformations sets a new direction for AER research, suggesting that TCNNs may offer a more streamlined and potentially more accurate approach to analyzing physiological signals.

We argue that while feature fusion approaches have their merits, incorporating TCNN could enhance their performance further. For example, replacing traditional CNNs in PETSFCNN with TCNNs might yield even higher accuracies. Furthermore, combining our TCNN model with multimodal data, such as GSR alongside ECG, could further improve the classification accuracy and robustness.

B. Modality

Emotion recognition hinges on both psychological and physiological components, with the latter being more accessible for computational analysis. Our study confirms that ECG and GSR are highly correlated with emotional states, aligning with findings from numerous studies. This high correlation makes ECG a practical choice for large-scale behavioral monitoring applications using wearable sensors.

In the literature, DL models for AER have focused predominantly on ECG data. Although multimodal approaches that integrate various physiological signals (e.g., EEG, EOG, and

EMG) have been explored, they have not consistently demonstrated substantial improvements over unimodal ECG-based models. This trend may be due to the strong predictive power of ECG signals alone or the current emphasis on showcasing the benefits of automated feature extraction over manual feature engineering. Future research should explore the potential of multimodal models more rigorously, particularly in real-world settings where different signals might provide complementary information.

C. Differentiating Emotions

The challenge of differentiating between specific emotions within the same arousal/valence quadrant remains an open problem. Most studies, including ours, have treated the classification problem as binary along each axis—arousal and valence. While our model successfully expands this to classify emotions along both axes simultaneously, differentiating between emotions, such as happiness and elation, which both fall within the same quadrant, requires a more nuanced approach.

We attempted to address this by training a regressor to predict the magnitude of arousal and valence, achieving a mean absolute error of 1.03. However, the resultant decrease in classification accuracy suggests that fine-grained differentiation may be more feasible in homogeneous datasets or with a more personalized approach. This limitation highlights the complexity of emotion recognition and the need for further research to refine these models for more precise emotion differentiation.

D. Other Applications and Future Work

The demonstrated effectiveness of TCNN for ECG signal analysis suggests its potential applicability to other time-series data. This approach could extend to other AER modalities, such as EEG and GSR, and even to nonemotional physiological analyses, such as monitoring periodontal bone loss or other health metrics.

Future work should explore inverse filtering for baseline removal to enhance ECG signal quality further. In addition, addressing motion artifacts through advanced preprocessing techniques or by integrating motion sensors can increase the robustness of AER models. Expanding datasets and standardizing preprocessing pipelines will be crucial for advancing the field and ensuring that improvements in model accuracy are not dataset-specific but generalizable across various conditions and populations.

In conclusion, while our TCNN model advances the state of the art in ECG-based AER, the field must continue evolving

through rigorous cross-modal research, improved preprocessing standards, and personalized emotion recognition models. By doing so, we can develop more accurate, reliable, and applicable emotion recognition systems for diverse real-world applications

VI. CONCLUSION

In this article, we have demonstrated the application TCNNs and residual block architectures for automated feature extraction of ECG signal data in AER problems. By leveraging the capabilities of TCNN, we were able to perform structural analysis during convolution, which streamlines the processing pipeline and avoids the need for separate convolutional and structural techniques that rely on feature fusion.

The methodologies and experiments conducted in this study highlight the potential of TCNN in maintaining the temporal dependencies of ECG signals, providing a robust framework for emotion classification. Our approach shows promise not only in improving the accuracy of AER systems but also in paving the way for future research to incorporate similar techniques in other biomedical signal processing applications.

This work sets the stage for further exploration into the integration of TCNN with multimodal data sources and the development of standardized preprocessing pipelines to enhance the reliability and comparability of AER models. By addressing these areas, we can move toward more accurate and generalizable emotion recognition systems that are applicable in various real-world contexts.

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